

Third Terminal Examination - 2082

Class : XI	Subject: Biology (Code: 2011)	Full Marks: 75
Time: 3 hrs.	SET 'A'	Pass Marks: 30

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt All Questions

PART : I [BOTANY]

Group 'A' [BOTANY]

Rewrite the best alternative of each question in your answer sheet. [6 × 1 = 6]

- A symbiotic relationship between a fungus & root of higher plant is called
 - Commensalism
 - Mutualism
 - Amensalism
 - Mycorrhizal association
- Which one of the following is responsible for the loss of fertility of soil?
 - Ammonification
 - Nitrification
 - Denitrification
 - Nitrogen assimilation
- Which one of the following occurs without the formation of spindle fibre and appearance of chromosome?
 - Amitosis
 - Mitosis
 - Meiosis
 - Metaphase
- Polymorphic organell having hydrolytic enzyme is
 - Lysosome
 - Golgi body
 - Mitochondria
 - Chloroplast
- What does the arc between C & A ($\widehat{CA}$) represent?
 - Tetradynamous
 - Epipetalous stamens
 - Ovary superior
 - C means corolla & A means androecium
- What types of food chain is the following?

Leaf Littor → Earthworm → frog → snake → hawk

 - Grazing food chain
 - Parasitic food chain
 - Predatory food chain
 - Detritus food chain

Group 'B' [BOTANY]

Give short answer to the following questions.

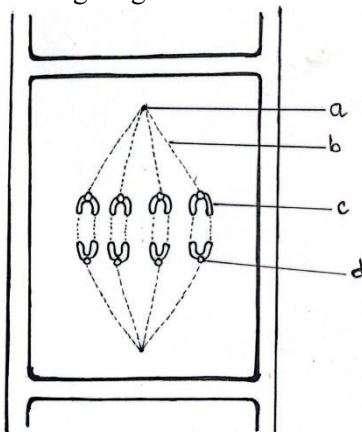
[4 × 4 = 16]

7. Define Mycology. Write down the general characters of fungi. [1+3]
8. Why gymnosperms are called naked seeded plants? Distinguish between angiosperms and gymnosperms. [1+3]
9. Describe asexual reproduction in *Mucor*. [4]

OR

Which cell organelle is called cytoplasmic skeleton? Write down the functions of Golgi body. [1+3]

10. Study the following diagram & answer the questions. [4]



- i. Label the parts a, b, c & d. [0.5×4]
- ii. Name the stage after it. [0.5]
- iii. Write down one most important character of this stage. [1]
- iv. Name the stage of cell division. [0.5]

Group 'C' [BOTANY]

Give long answer to the following questions.

[8 × 2 = 16]

11. Which two processes help to balance carbon cycle in nature? What is the percentage of CO₂ in atmosphere? What is the effect of increased concentration of CO₂ in atmosphere? Describe carbon cycle in detail. [1+1+1+5]
12. Describe the family Fabaceae in semi technical terms with floral formula and floral diagram. Also state the scientific names of any two economic plants of the family. [3+4+1]

OR

Why pteridophytes are called vascular cryptogams? Describe the life cycle of *Dryopteris*. [1+7]

PART : II [ZOOLOGY]

Group 'A' [ZOOLOGY]

Rewrite the best alternative of each question in your answer sheet. [5 × 1 = 5]

13. Which stage of *Plasmodium* is taken up by the mosquito during a blood meal?
 - a. Sporozoite
 - b. Gametocyte
 - c. Merozoite
 - d. Trophozoite
14. In earthworms, excretion is carried out by which type of structures?
 - a. Nephridia
 - b. Malpighian tubules
 - c. Flame cells
 - d. Kidneys
15. Who proposed the theory of inheritance of acquired characters, which suggested that traits developed during an organism's lifetime could be passed on to offspring?
 - a. Charles Darwin
 - b. Jean-Baptiste Lamarck
 - c. Alfred Russel Wallace
 - d. Gregor Mendel
16. Who conducted the famous experiment in 1953 that simulated early Earth conditions and demonstrated the formation of amino acids from simple molecules?
 - a. Louis Pasteur
 - b. Stanley Miller and Harold Urey
 - c. Alexander Oparin
 - d. Charles Darwin
17. The tympanum in frog functions as:
 - a. Respiratory organ
 - b. Hearing organ
 - c. Excretory organ
 - d. Digestive organ

Group 'B' [ZOOLOGY]

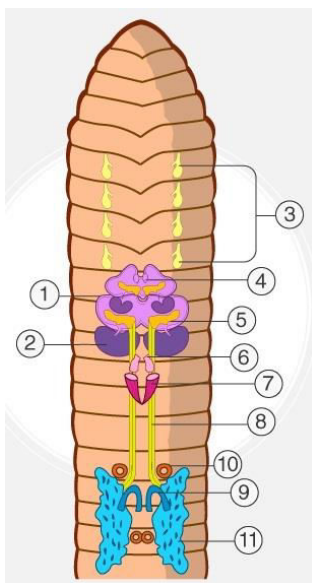
Give short answers to the following questions. [4 × 4 =16]

18. List any 6 major functions of liver of frog. [4]
19. What type of reproduction is schizogony? Describe the process of liver schizogony in *Plasmodium vivax* with labeled diagram. [1+3]
20. Describe the habit and habitat of frog. [4]

21. Observe the figure given below and answer the questions.

[2+1+1]

- Label 3, 4, 5 and 6.
- In which segments label 10 and 11 are present?
- Mention the function of 3.



Group 'C' [ZOOLOGY]

Give long answers to the following questions.

[2 × 8 = 16]

22. What is excretion? Name the different types of nephridia present in earthworm's body with labeled diagram discuss the structure of septal nephridia.

[1+1+4+2]

OR

Draw well labelled diagram of alimentary canal of frog. Describe the major parts of alimentary canal.

[2+6]

23. Define evolution? How neo-Darwinism explains the evolution of life in nature?

[1+7]

ॐ The End ॐ